



**Meridian Junior College**  
**JC2 Preliminary**  
**Examinations**

**Name:**

**Class:**

**Index No:**

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# **BIOLOGY**

## **9747/01**

Paper 1 Multiple Choice

18 Sep 2008

1 hour 15 minutes

Additional Materials: OMR answer sheet

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### **READ THESE INSTRUCTIONS FIRST**

**Do not open this booklet until you are told to do so.**

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, civics group and register number in the spaces at the top of this page.

There are **forty** questions in this section. Answer **all** questions. For each question, there are four possible answers labelled **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate OMR answer sheet.

**Read the instructions on the Answer Sheet very carefully.**

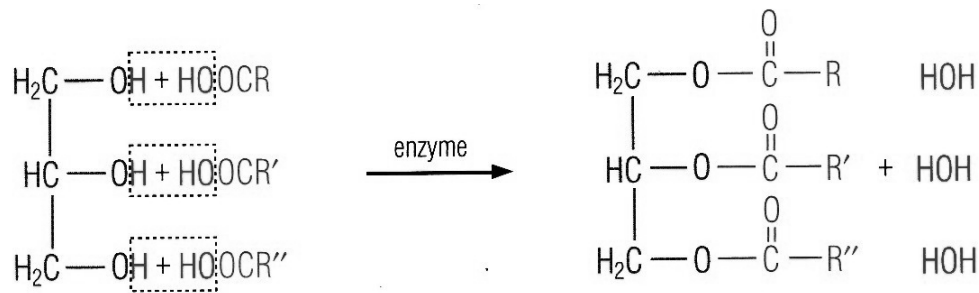
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Calculators may be used.

1. The helical form of a polypeptide chain is due to the presence within the molecule of
- Covalent bonds
  - Disulphide bonds
  - Peptide bonds
  - Hydrogen bonds**

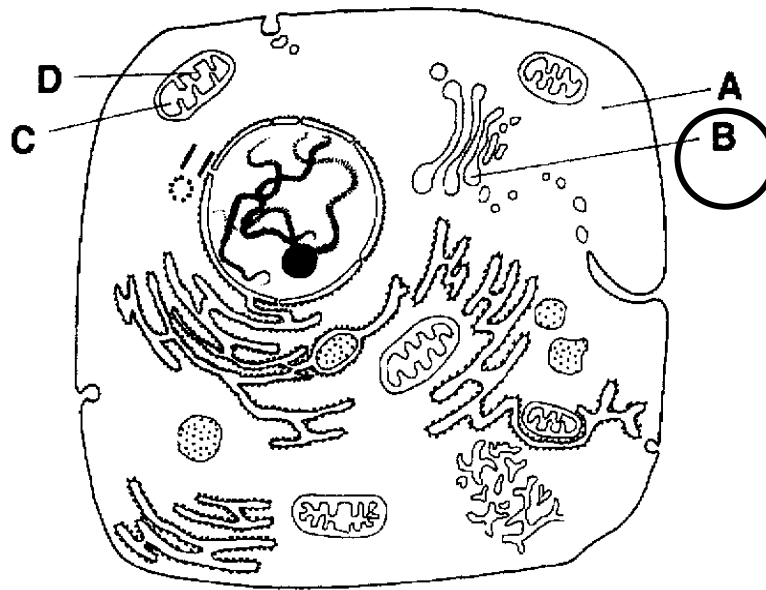
2. The diagram below shows the formation of a



- Fatty acid
  - Triglyceride**
  - Phospholipid
  - Ester linkage
3. Which of the following is absent in prokaryotic cells but present in eukaryotic cells?
- cell wall
  - nucleus**
  - flagellum
  - organelles
4. Some cellular organelles are bounded by a single membrane, while other organelles have two membranes (envelopes) around them. Which one of the following is correct?

|           | Single membrane  |                 | Two membranes  |                    |
|-----------|------------------|-----------------|----------------|--------------------|
| <b>A.</b> | <b>tonoplast</b> | <b>lysosome</b> | <b>nucleus</b> | <b>chloroplast</b> |
| B.        | chloroplast      | lysosome        | nucleus        | tonoplast          |
| C.        | nucleus          | chloroplast     | lysosome       | tonoplast          |
| D.        | tonoplast        | chloroplast     | nucleus        | lysosome           |

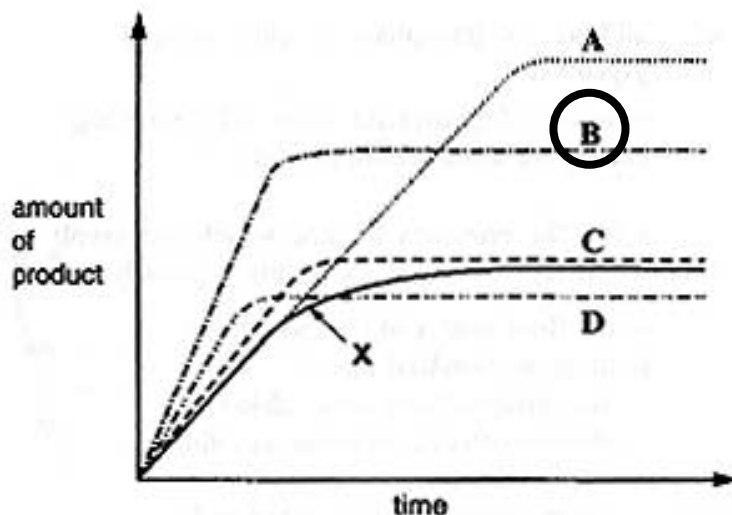
5. At which site is the addition of an oligosaccharide to a protein possible?



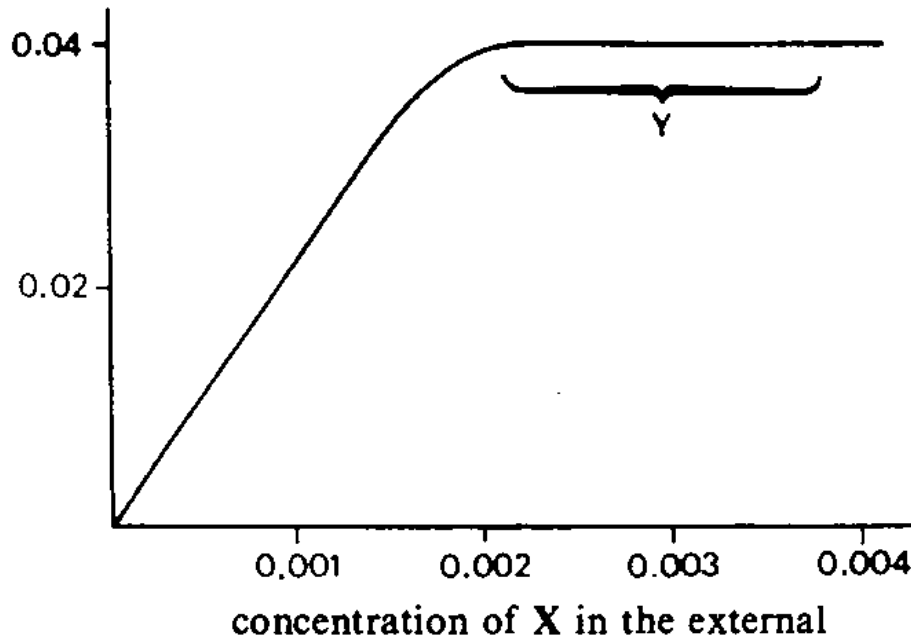
6. A series of enzymes catalyze the reaction  $X \rightarrow Y \rightarrow Z \rightarrow A$ . Product A binds to the enzyme that converts X to Y at a position remote from its active site. This binding decreases the activity of the enzyme. Product A is

- A. a coenzyme.
- B. an allosteric inhibitor.**
- C. a substrate.
- D. a competitive inhibitor.

7. Line X shows the activity of an enzyme at 20°C. Lines A to D show the effect of different conditions on the activity of the enzyme. Which line shows the effect of adding extra substrate provided enzyme concentration is not limiting?



8. Substance X (a mineral ion) is actively transported into cells. Equal-sized samples of cells were placed in media containing different concentrations of X for an hour. The intracellular concentration of X was then measured. All other metabolic conditions were maintained at the optimum level. The graph below shows the results.



From the information given above, which one of the following would account for the level region of the graph labelled Y?

- A. A respiratory inhibitor had been introduced.
  - B. All the active transport carriers had been operating at their maximum rate.**
  - C. The active transport carriers had been inactivated by a non-competitive inhibitor.
  - D. As the internal concentration of X rose, more of the substance X was metabolised.
9. If photosynthesizing green algae are provided with  $\text{CO}_2$  synthesized with heavy oxygen ( $^{18}\text{O}$ ), later analysis will show that all but one of the following compounds produced by the algae contain the  $^{18}\text{O}$  label. That one exception is
- A. PGA
  - B. Glucose
  - C. RuBP
  - D.  $\text{O}_2$**

10. Some photosynthetic organisms contain chloroplasts that lack photosystem II, yet are able to survive. The best way to detect the lack of photosystem II in these organisms would be

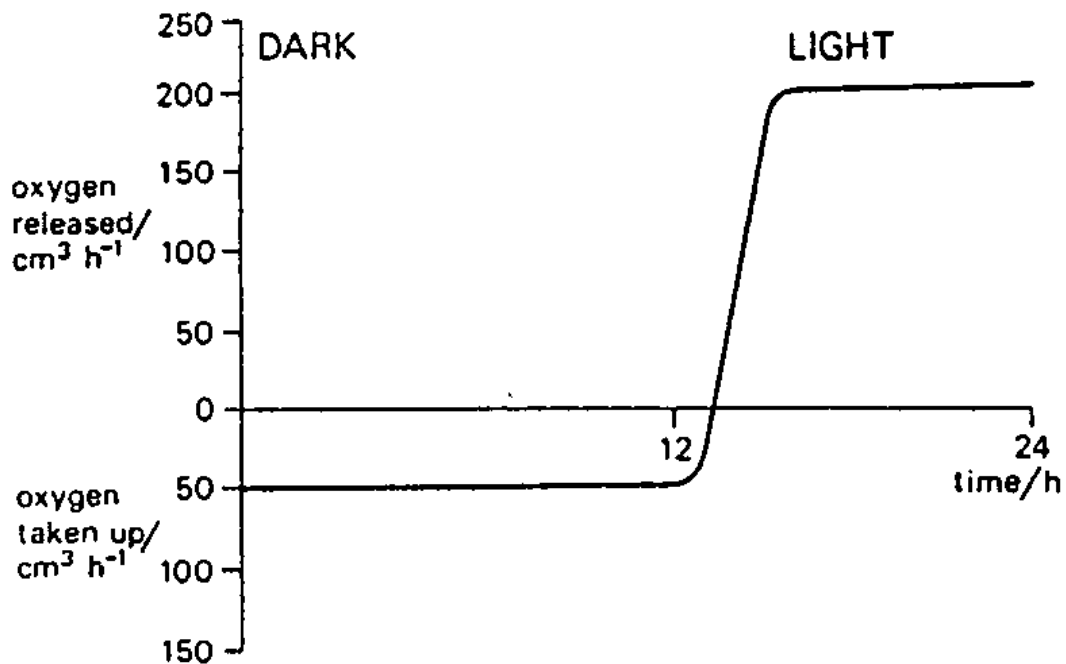
A. to determine if they have thylakoids in the chloroplasts.

**B. to test for liberation of  $O_2$  in the light.**

C. to test for  $CO_2$  fixation in the dark.

D. to test for production of either sucrose or starch.

11. Data were obtained relating to the rates of oxygen release and uptake in plants. The plants were placed for 12 hours in the dark followed by 12 hours in the light. Temperature was constant throughout the experiment. The results are shown in the graph below.



Which one of the following is the most accurate estimate of the total volume of oxygen used by the plant for respiration during the 24 hours of the experiment?

A.  $50\text{ cm}^3$

B.  $600\text{ cm}^3$

**C.  $1200\text{ cm}^3$**

D.  $1800\text{ cm}^3$

12. A major function of the mitochondrial inner membrane is the conversion of energy from electrons to the stored energy of the phosphate bond in ATP. To accomplish this function, this membrane must have all of the following features except
- A. proteins to accept electrons from NADH.
  - B. integral, transverse ATP synthase.
  - C. proton pumps embedded in the membrane.
  - D. high permeability to protons.**
13. Which of the following is true of synaptic signalling and hormone signalling?
- A. Hormone signalling occurs in animals only whereas synaptic signalling occurs in plants only.
  - B. Hormone signalling is more important between cells that are at greater distances apart than in synaptic signalling.**
  - C. Both use neurotransmitters, but hormone signalling is for adjacent cells in animals only.
  - D. Both are forms of paracrine signalling.
14. The activity of a protein regulated by phosphorylation
- A. depends mostly on the concentration of inorganic phosphate inside the cell.
  - B. depends on the balance in the cell between active kinase and active phosphatase enzymes.**
  - C. is dependent on the site of attachment of the protein to the plasma membrane.
  - D. is independent of kinase activity, but directly dependent on phosphatase activity inside the cell.
15. The vertebrate system conducts impulses in only one direction. This one-way conductance across synapse occurs
- A. because the  $\text{Na}^+ - \text{K}^+$  pump moves ions in one direction.
  - B. as a result of voltage-gated sodium channels found in the vertebrate system.
  - C. because only the postsynaptic membranes can bind neurotransmitters.**
  - D. because vertebrate nerve cells have dendrites.

16. Certain drugs act at synapses and affect the action of neurotransmitter substances. The table shows the effects of four different drugs.

| Drug | Effect                                        |
|------|-----------------------------------------------|
| 1    | Inhibits the enzyme cholinesterase            |
| 2    | Prevents the release of acetylcholine         |
| 3    | Competes with acetylcholine at receptor sites |
| 4    | Stimulates the release of acetylcholine       |

Which two drugs would prevent a skeletal muscle from responding to an electrical stimulus in the pre-synaptic neurone?

A. 1 and 2

**B. 2 and 3**

C. 1 and 4

D. 3 and 4

17. If protein was the only food available to a dog and the supply of protein was only half the minimum required to supply its energy needs, the dog would show an increase in the

**A. Hydrolysis of glycogen**

B. Synthesis of proteins

C. Synthesis of glycogen

D. Concentration of amino acids in the blood above normal level

18. Blood samples taken from an individual who had been fasting for 24 hours would have which of the following?

|   |                         |
|---|-------------------------|
| 1 | high levels of insulin  |
| 2 | high levels of glucagon |
| 3 | low levels of insulin   |
| 4 | low levels of glucagon  |

A. 1 and 2

**B. 2 and 3**

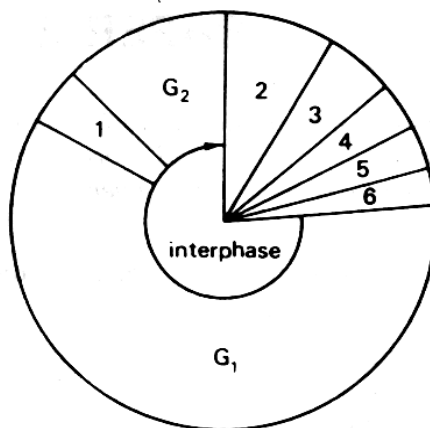
C. 3 and 4

D. 1 and 4

19. Which option most correctly describes the type of cell division occurring in the following organs?

|    | Ovary of a rat             | Testis of a human          | Brain of a human    | Root of a plant     |
|----|----------------------------|----------------------------|---------------------|---------------------|
| A. | Meiosis only               | Meiosis only               | Mitosis and meiosis | Mitosis and meiosis |
| B. | Meiosis only               | Mitosis only               | Mitosis only        | Meiosis only        |
| C. | Mitosis only               | Mitosis and meiosis        | Mitosis only        | Mitosis only        |
| D. | <b>Mitosis and meiosis</b> | <b>Mitosis and meiosis</b> | <b>Mitosis only</b> | <b>Mitosis only</b> |

20. The diagram below shows the cell cycle. G<sub>1</sub> and G<sub>2</sub> represent periods of intensive synthesis within the cytoplasm.



Which of the following correctly identifies the other cellular events associated with mitosis?

|    | 1                    | 2               | 3                | 4               | 5                | 6                  |
|----|----------------------|-----------------|------------------|-----------------|------------------|--------------------|
| A. | cytokinesis          | DNA synthesis   | prophase         | telophase       | metaphase        | anaphase           |
| B. | telophase            | cytokinesis     | DNA synthesis    | prophase        | metaphase        | anaphase           |
| C. | <b>DNA synthesis</b> | <b>prophase</b> | <b>metaphase</b> | <b>anaphase</b> | <b>telophase</b> | <b>cytokinesis</b> |
| D. | prophase             | metaphase       | anaphase         | telophase       | DNA synthesis    | cytokinesis        |



21. A diploid cell contains four chromosomes ( $2n = 4$ ). If this cell divides by mitosis, and there is no mutation, how many genotype(s) exist among the daughter cells?
- A. 1  
B. 2  
C. 4  
D. 8
22. The DNA code for glutamic acid is CTC or CTT. The code for valine is CAA or CAT. In sickle cell haemoglobin, valine is present instead of glutamic acid. Assuming a single base pair substitution has occurred, what is the mRNA code in the affected mutant?
- A. CUU  
B. GAA  
C. GUU  
D. **GUA**
23. Choose the answer that has these events of protein synthesis in the proper sequence.
1. An aminoacyl-tRNA binds to the A site.
  2. A peptide bond forms between the new amino acid and a polypeptide chain.
  3. Protein release factor binds to A site.
  4. A small ribosomal subunit binds with mRNA.
  5. tRNA translocates to the P site from the A site.
- A. 1, 3, 2, 4, 5  
B. **4, 1, 2, 5, 3**  
C. 4, 1, 3, 2, 5  
D. 2, 4, 5, 1, 3
24. If proteins were composed of only 12 different kinds of amino acids, what would be the smallest possible codon size in a genetic system with four different nucleotides?
- A. 1  
B. **2**  
C. 4  
D. 12

25. Which of the following does not contribute to the emergence of viral disease?

- A. production of new strains of virus through mutation
- B. spread of existing virus from one host to another
- C. switch from lytic to lysogenic activity**
- D. dissemination from a small, isolated host population

26. Which of the following correctly describes the characteristics of a virus?

|           | Obligate parasitism | Involvement of host enzymes | Involvement of host cell ribosomes | Entry of nucleic acid into a cell |
|-----------|---------------------|-----------------------------|------------------------------------|-----------------------------------|
| A.        | √                   | X                           | √                                  | X                                 |
| B.        | √                   | X                           | √                                  | √                                 |
| C.        | √                   | √                           | X                                  | X                                 |
| <b>D.</b> | √                   | √                           | √                                  | √                                 |

27. Bacterial Strain A has been infected with viruses. Upon lysis, the virions produced are introduced to Bacterial Strain B. After a short period of time, a new strain of bacteria is detected that is very similar to Strain A but has a few characteristics of Strain B. Which is the process that leads to the production of the new strain?

- A. transduction**
- B. transposition
- C. transformation
- D. conjugation

28. A eukaryotic gene typically has all of the following features except

- A. introns.
- B. a promoter.
- C. an operator.**
- D. control elements.

29. Which of the following event(s) is/are necessary for the production of a malignant tumour?

1. activation of an oncogene in the cell
2. the inactivation of tumour-suppressor genes within the cell
3. the presence of mutagenic substances within the cell's environment
4. the presence of a retrovirus within the cell

**A. 1 and 2**

B. 1 only

C. 3 only

D. 3 and 4

30. In mice, the gene for 'dappled' coat (D) and its recessive allele, the gene for 'plain' coat (d), are located on the X chromosome. The gene for 'straight' whiskers (W) and its recessive allele, the gene for 'bent' whiskers (w), are autosomal. A male mouse with plain coat and bent whiskers was mated on several occasions to the same female and the large number of offspring consisted of the following phenotypes in equal proportion:

- dappled male with straight whiskers
- dappled female with straight whiskers
- dappled male with bent whiskers
- dappled female with bent whiskers
- plain male with straight whiskers
- plain female with straight whiskers
- plain male with bent whiskers
- plain female with bent whiskers

If  $X^D$  represents an X chromosome carrying a gene for 'dappled' coat and  $X^d$  represents an X chromosome carrying a gene for 'plain' coat, what is the genotype of the female parent?

A.  $X^D X^D WW$

B.  $X^D X^D Ww$

C.  $X^D X^d WW$

**D.  $X^D X^d Ww$**

31. Given the parents AABBCc X AabbCc, assume simple dominance and independent assortment. What proportion of the progeny will be expected to phenotypically resemble the first parent?

- A.  $\frac{1}{4}$
- B.  $\frac{1}{8}$
- C.  $\frac{3}{4}$
- D.  $\frac{3}{8}$

32. In garden peas, the allele T code for terminal position of flowers and is dominant over the allele t, which codes for axil position. In an experiment, 2 garden pea plants, both heterozygous for terminal flowers, produced 77 offspring with flowers at the terminal position and 43 offspring with flowers at the axil position. A  $\chi^2$  test was done to determine whether the results of the experiment follow the expected pattern of inheritance. Given that the  $\chi^2$  critical value at 1 degree of freedom and a probability of 0.05 is 3.84, which of the following conclusions can be drawn from the experiment and the  $\chi^2$  test?

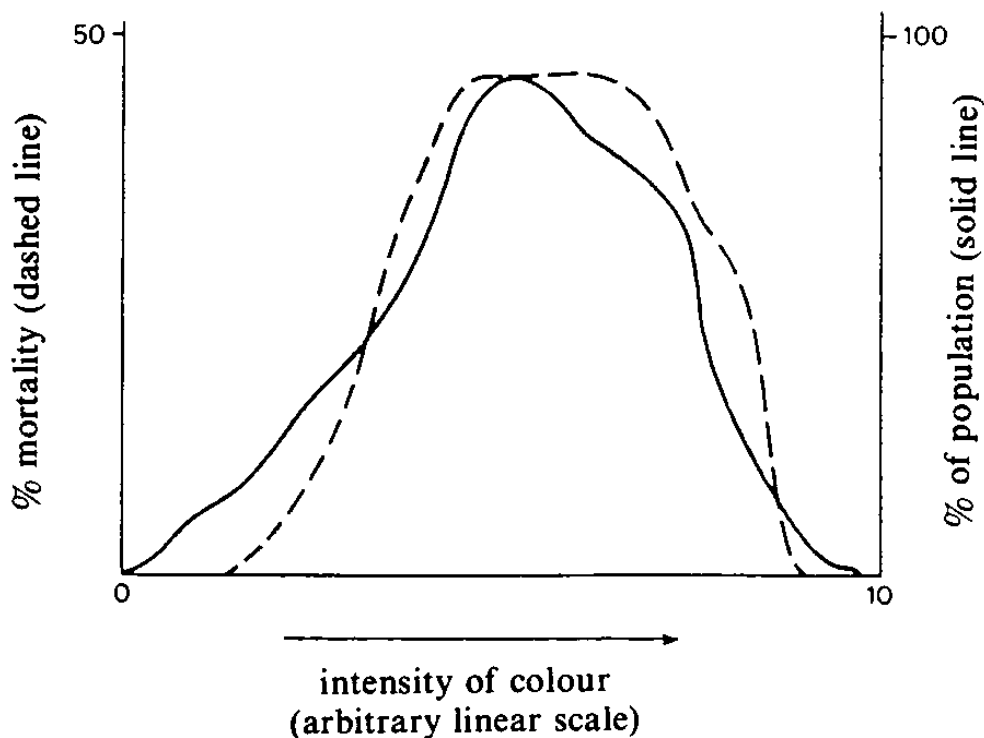
$$\text{Note: } \chi^2 = \sum \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}} = \sum \frac{(O - E)^2}{E}$$

- A. **Since the  $\chi^2$  value obtained from the experiment is greater than 3.84, flower position in garden peas does not involve a single pair of segregating alleles at the T locus.**
- B. Since the  $\chi^2$  value obtained from the experiment is greater than 3.84, flower position in garden peas involves a single pair of segregating alleles at the T locus.
- C. Since the  $\chi^2$  value obtained from the experiment is less than 3.84, flower position in garden peas does not involve a single pair of segregating alleles at the T locus.
- D. Since the  $\chi^2$  value obtained from the experiment is less than 3.84, flower position in garden peas involves a single pair of segregating alleles at the T locus.

33. What would be the best technique for determining the evolutionary relationships among several closely related species, each of which still contains living members?

- A. examining the fossil record
- B. comparison of homologous structures
- C. comparative embryology
- D. DNA or RNA analysis**

34. The graphs below were drawn from data on a population of a species of moth which shows considerable variation in colour intensity.



Which one of the following can be correctly concluded from these graphs?

- A. Colour variations are environmentally induced.
- B. Extreme forms are positively selected.**
- C. Intermediate forms are positively selected.
- D. The species shows discontinuous variation with respect to colour.

35. One finds that organisms on islands are different from, but closely related to, similar forms found on the nearest continent. This is taken as evidence that

- A. island forms and mainland forms descended from common ancestors.**
- B. common environments are inhabited by the same organisms.
- C. the island forms and mainland forms are converging.
- D. island forms and mainland forms have identical gene pools.

36. Assume that you are trying to insert a gene into a plasmid and someone gives you a preparation of DNA cut with restriction enzyme X. The gene you wish to insert has sites on both ends for cutting by restriction enzyme Y. You have a plasmid with a single site for Y, but not for X. Your strategy should be to
- A. insert the fragments cut with X directly into the plasmid without cutting the plasmid.
  - B. cut the plasmid with restriction enzyme X and insert the fragments cut with Y into the plasmid.
  - C. cut the DNA again with restriction enzyme Y and insert these fragments into the plasmid cut with the same enzyme.**
  - D. cut the plasmid twice with restriction enzyme Y and ligate the two fragments onto the ends of the human DNA fragments cut with restriction enzyme X.
37. Bacteria containing recombinant plasmids are often identified by which process?
- A. examining the cells with an electron microscope
  - B. using radioactive tracers to locate the plasmids
  - C. exposing the bacteria to an antibiotic that kills the cells lacking the plasmid**
  - D. producing antibodies specific for each bacterium containing a recombinant plasmid
38. Which of the following procedures would produce RFLPs?
- A. incubating a mixture of single-stranded DNA from two closely related species
  - B. incubating DNA nucleotides with DNA polymerase
  - C. incubating DNA with restriction enzymes**
  - D. incubating RNA with DNA nucleotides and reverse transcriptase
39. Which of the following correctly describes the uses of the different types of tissue culture?

|           | Tissue culture using | Uses                         |
|-----------|----------------------|------------------------------|
| <b>A.</b> | <b>anthers</b>       | <b>Obtain haploid plants</b> |
| B.        | protoplasts          | Obtain haploid plants        |
| C.        | shoot tips           | Produce hybrids              |
| D.        | callus               | Produce hybrids              |

40. The following are advantages of transgenics except:

- A. A gene for a desirable characteristic can be identified and cloned.
- B. All the beneficial characteristics of an existing variety can be kept and just the desired new gene can be added.
- C. Sexual reproduction is necessary.**
- D. Transgenics is much faster than conventional breeding.